

DSAIT4335 Recommender Systems

Final Project

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Feedback session

- Feedback sessions for final project in week 6
 - October 7th and 8th
- The duration is 20 minutes per group
- Enroll for timeslot in:
<https://queue.tudelft.nl/lab/9749>
- Location: Building 28, 1st floor, room Hardy (1.W950)

Final Project

- **LO 5: Implement a recommendation algorithm to operate in a specific domain**
 - Implement hybrid recommendation models
- **LO 6: Evaluate the effectiveness of a recommendation model through an offline evaluation**
 - Run offline experiments with each individual and hybrid recommendation models and evaluate the performance
 - Compare the performance of each with baselines
 - Coefficient analysis: impact of each recommendation model on successful recommendation in a hybrid setting
- **LO7: Analyze the recommendation results with respect to known challenges and societal aspects**
 - Performance analysis beyond accuracy: diversity, novelty, calibration, fairness and popularity bias

Experimental Setup

- Use RecBole for implementing your project
 - Pull the repository to get the updated codes with configurations of more algorithms
 - You have access to the configuration files for each model to tune it
 - In each experiment, RecBole returns intermediate files like train/val/test sets as well as recommendation results
 - Perform the evaluation separately
- The focus will be on ranking task
 - RecBole configurations are prepared accordingly
- Run experiments on MovieLens 100K dataset

Task 1: Hybrid recommender

Building individual recommenders

- **Task 1.1)** Build individual recommendation models
 - The models discussed in the class (except content-based, all are implemented in RecBole)
- **Task 1.2)** Tune the individual recommender models
- **Task 1.3)** Build weighted hybrid recommender model
 - Learn the coefficient for individual recommenders using a **regression model**
- **Task 1.4)** Build other hybrid models discussed in the class
 - What other hybrid recommender models are interesting to look at?
- **Task 1.5)** Tune your hybrid recommender models

Task 2: Evaluation of Effectiveness

- **Task 2.1)** Implement the evaluation metrics
 - Both accuracy and beyond-accuracy metrics
 - **Note:** While some metrics are already implemented in RecBole, implement all metrics yourself separately for consistency of your evaluation
 - Specifically for beyond-accuracy metrics that are not included in RecBole
- **Task 2.2)** Run experiments with the best-performing models
 - Compare the performance
 - Discuss the models' performance with respect to naïve baselines, i.e., random and popularity models

Task 2: Evaluation of Effectiveness

- **Task 2.3)** Analyze your hybrid models
 - Conduct coefficient analysis
 - How does each individual recommender contribute to the final prediction of the hybrid recommender?
- **Task 2.4)** Think about designing insightful analysis
 - What types of analysis can be designed to reason/explain the behavior of hybrid recommender or individual models within the hybrid model?

Task 2: Evaluation of Effectiveness

- **Task 2.5)** Analyze the models' performance (both accuracy and beyond-accuracy) with respect to different user and item groups
 - Where is each recommendation model successful in delivering accurate recommendation?
 - For which user and item groups each recommendation model is more successful?
- **Task 2.6)** What insights can you gain from this analysis?
 - Does it suggest or lead to the development of novel and superior hybrid recommendation models?

Task 3: Societal Aspects

- **Task 3.1)** Implement rerankers improving the beyond accuracy of the recommendation models
 - Diversification method, calibration method, fairness method (user-side and item-side)
- **Task 3.2)** Evaluate the performance of recommendation models after applying the rerankers
 - Discuss the beyond-accuracy performance
 - Discuss the trade-off between accuracy and beyond accuracy metrics

Task 3: Societal Aspects

- **Task 3.3)** Evaluate your hybrid recommender models when built by
 - 1) Individual recommender + reranker
 - 2) Individual recommenders and then applying rerankers afterwards
 - Discuss your observations
- **Task 3.4)** Analyze the impact of rerankers on the quality of recommendations for groups of users and items

Deliverables

Report

Codes

Peer evaluation

Deliverables

- Group submission (one per group)
 - Report -> pdf file
 - Codes -> directory containing runnable code files
- Individual submission (one per student)
 - Peer evaluation -> Excel file
- For group submission
 - Make a .zip file including report and code files
 - Name the .zip file as your group number, e.g., for group 30, it would be “30.zip”
- Submit the files to Brightspace
 - Deadline is **October 26th, 23:59**

Report

- The report includes the presentation of experimental results and your reflection on the tasks
- Structure
 - First page: the group information, e.g., students' name and group number
 - Main body: experimental results and corresponding discussion/reflection for each task
 - Maximum one page for each task, containing both experimental results and discussion
 - Maximum 200 words for the discussion of experimental results for each task
 - Appendix: presenting additional experimental results
 - Same criteria in main body applies here

Report

- The main body of your report should be self-contained
 - Report the most important and insightful parts of your experimental results
 - Appendix will not be counted if the main body is lacking the essential experimental results and observations
- Formatting guideline
 - Times New Roman
 - Font size 12
 - Line spacing 1.15

Report

- Be clear on your experimental designs/implementations
 - Explain clearly your assumptions, choices, etc
- Be creative in organizing your description (design choices + discussion) for each task

Report

- Depth of analysis is more important than the number of models or metrics you include in your experiments
- Avoid comparing the numbers and making conclusion
 - E.g., nDCG of model A is higher than model B, then model A is better
- Instead, discuss and compare with respect to the characteristics of the models
 - E.g., model A yielded higher nDCG because it better handles sparsity issue according to evidence in Figure 1

Codes

- Place all your code files in a directory
- The code should be runnable
 - Include a short instruction how running your code, e.g.,
readme.md file
- Prepare the code in a clean and understandable manner

Peer evaluation

- You will evaluate your teammates from different aspects including contribution, responsibility, communication, etc
- Your final grade will be affected if your teammates confirm that you did not perform well in the group

Next session ...

- Feedback session on assignments 1, 2, and 3
 - A chance for you to clarify the questions you might have
 - Specifically, for those who have difficulty running RecBole